

Use-Case Flow Specification

UC-01: Submit Damage Report

Municipal Infrastructure Damage Reporting App — Problem Statement #29

Use Case

UC-01: Submit Damage Report

Actors

Primary Actor: Citizen Reporter

Supporting Actors: GIS Mapping Service (extracts/validates location), Municipal Engineer (receives resulting ticket)

Preconditions

- The citizen has the mobile/web app installed and is logged in (or using guest mode with a valid contact number).
- The device camera and location/photo-library permissions have been granted to the app.
- The municipal GIS map and department-routing rules are available to the system.

Postconditions

- A new damage report ticket exists in the system with a unique Ticket ID, a damage category, at least one geotagged photo, and a location.
- The ticket has been automatically dispatched to the correct municipal engineering department queue.
- The citizen can view the ticket's status via Track Report Status (UC-07).

Main Success Scenario

1. Citizen selects "Report Damage" from the app home screen.
2. System displays a form prompting the citizen to select a damage category: Pothole, Broken Streetlight, or Water Pipeline Leak.
3. Citizen selects a category and captures or uploads at least one photo of the damage.
4. System extracts EXIF GPS metadata from the photo (includes UC-02: Extract GPS Location) and pins the incident on the municipal GIS map.
5. System displays the detected location and a short description field to the citizen for confirmation.
6. Citizen reviews the auto-filled location, optionally adds a text description, and taps "Submit Report."
7. System validates that a category and at least one geotagged photo are present.
8. System generates a unique Ticket ID and timestamps the report.
9. System automatically routes the ticket to the municipal engineering department mapped to the damage category (includes UC-06: Route Ticket to Department).
10. System displays a confirmation screen to the citizen showing the Ticket ID and initial status "Submitted."
11. Use case ends successfully.

Alternate Flow

4a. GPS Metadata Missing or Invalid

- 4a1. System detects that the uploaded photo has no usable EXIF GPS metadata (e.g., GPS was disabled when the photo was taken).
- 4a2. System prompts the citizen to manually pin the incident location on the GIS map (extends UC-02 with UC-03: Manually Pin Location on Map).
- 4a3. Citizen drags the map pin to the correct location and confirms.
- 4a4. Flow resumes at step 5 of the Main Success Scenario using the manually pinned coordinates.

7a. Required Fields Missing

- 7a1. System detects that no damage category was selected or no photo was attached.
- 7a2. System displays a validation message identifying the missing field(s) and blocks submission.
- 7a3. Citizen supplies the missing information and re-attempts submission at step 6.

Special Requirements

- Related to FR-001 (GPS extraction), FR-002 (category/photo capture), FR-003 (automated routing) and NFR-002 (encrypted transit/storage of photo and location data).